Step 8 summary - mock CSC662 paper vs 3 clean modern papers

Scope. mock_paper_01.pdf (5 pages) vs JUN 2019 (7), DEC 2019 (6), FEB 2022 (14). Word COM export; 200 dpi renders. Full report: step8_comparison.md. Slide check: slide_convention_report.md.

Structure

Feature	Mock	JUN 2019	DEC 2019	FEB 2022
Part A / B	10 x (2+6) = 80 / 2 x (5+5) = 20	same	same	same
Total / duration	100 / 3 HOURS	100 / 3 HOURS	100 / 3 HOURS	100 / 3 HOURS
Part headings	PART A / PART B	PART A (80 MARKS) / PART B (20 MARKS)	same	same
Cover	FINAL EXAMINATION, 5 instructions	same style	same style	FINAL TEST, 3 instructions + name/marks grid
Printed pages	5	7	6	14
Computed / placeholder marks	28 / 72	6 / 0	6 / 0	0 / 0

Matches: parts, question counts, 2/6/5/5 marks, 100 total, 3 hours, a/b numbering, header/footer on every body page, right-aligned mark labels.
Differences: mock part headings omit mark totals; cover differs from FEB 2022; page count/density; computed share.

Layout (pages 2+)

	Mock	JUN 2019	DEC 2019	FEB 2022
Header / footer present	4/4 / 4/4	6/6 / 6/6	5/5 / 5/5	13/13 / 13/13
Header / footer / body ink (median)	0.0116 / 0.0072 / 0.0270	0.0038 / 0.0102 / 0.0368	0.0116 / 0.0105 / 0.0459	0.0125 / 0.0166 / 0.0184
Question heading	10.7 pt regular, x=0.123	OCR only	9.7-10.0 regular, x=0.136	11.0 bold, x=0.121
a/b labels / marks labels	24 / 24	24 / 24 (OCR)	24 / 24	25 / 24
Mark right edge / page width	0.879	OCR only	0.868	0.879

Mock is denser than all three (4 questions/page on pages 2-4; Part B Q2 spills onto p5; FEB 2022 gives each question a page), footer ink is lighter than DEC/FEB, and FEB 2022 uses bold headings. Side-by-side question pages: step8_side_by_side.png (DEC), _jun2019.png, _feb2022.png.

Computed share

Clean modern fingerprint: **40 machine-verifiable marks over 10 papers = 4.0 marks/paper** (34 classical cipher + 6 ACM construction). The three clean papers carry 6, 6 and 0. The mock carries **28 marks (6 slots)** and **72 placeholder marks (18 slots)** - 7x the modern computed share. **Mock is a practice-mode paper.** A realistic paper would have about 2-5 computed marks (one 6-mark cipher/ACM question, sometimes none) and 94-98 theory marks; the 72 placeholders are the missing realistic part.

Wording and slide verification

Stems now state the key/method: Caesar shift 7 with direction; Playfair 5 x 5 grid, I/J merge, X filler; columnar 3 x 4 matrix, put ciphertext into columns/read rows, filler X.

- **Caesar (Q2a/Q4a):** 24-25 letters, shift 7, 2 marks; real modern Caesar is 11 letters, 6 marks, asks for the key; older 15 letters, 3 marks.
- **Playfair (Q2b/Q4b):** 12-13 pairs, key REDFOX, 6 marks vs real 10 letters/5 pairs; **no lecture slide**; I/J + X unverified.
- **ACM (Q3b):** matrix+ACL+CL+diagram for 6 marks; older three views 3+3+3 = 9 marks; modern matrix only, 6 marks.
- **Columnar (Q5b):** 3 x 4, 12 letters, pad X, 6 marks; **matches lecture slides 20-21 and 24**; real questions use 30 letters, pad D/z and "Show the steps".
- **Other slide checks:** Caesar (5/6/9), Vigenere (15), rail fence (25/26) match. **Playfair/affine: no slide. OTP slide 17 is internally inconsistent** (I/J-merged plaintext indices, 26-letter ciphertext, Q for 15); OTP remains unverified.

Marking scheme

mock_scheme_01.pdf, 5 pages, same header/footer as the paper: working for all 6 computed slots (Caesar arithmetic, Playfair grid+pairs, ACM matrix/ACL/CL + rendered diagram, 3x4 columnar grid), confidence labels (computed / convention unverified / placeholder), suggested mark splits, and an 18-row placeholder table. No official CSC662 schemes exist; all splits are inferred from the printed 2/6 and 5/5 labels. Columnar and Caesar now carry slide evidence; Playfair remains unverified.

Weakness list (specific)

- 72 marks are placeholders; the theory module is not built.
- Computed share 28 % vs 4.0 marks/paper modern average (practice mode).
- Caesar twice and Playfair twice; real papers vary cipher types.
- Caesar 24-25 letters for 2 marks; Playfair 12-13 pairs for 6 marks and no slide.
- ACM four outputs for 6 marks; older three-view question is 9 marks.
- Columnar 12 vs 30 letters; 3x4 convention verified, but real questions use pad D/z.
- OTP slide inconsistent; Playfair/affine have no slide; single transposition is paper-only.
- Part headings omit "(80 MARKS)/(20 MARKS)"; cover differs from FEB 2022.
- 5 pages vs 6/7/14; less answer space; headings regular vs FEB bold; footer ink lighter.
- No official marking scheme; suggested splits are inferred.